

Caio Lins Peixoto

MSc Student

Rio de Janeiro

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Education

- 2024 — Now **MSc in Applied Mathematics**, FGV EMAP, Brazil – RJ, 4.0/4.0.
Working on infinite dimensional optimization, stochastic control and nonparametric statistics.
- 2020 — 23 **BSc in Applied Mathematics**, FGV EMAP, Brazil – RJ, 9.8/10.
Thesis: *Nonparametric Instrumental Variable Regression through Stochastic Gradients and Kernel Methods*.

Work Experience

Research

- 2023—24 **Research Assistant** Co-authored the paper “*Nonparametric instrumental variable regression through stochastic approximate gradients*”. I developed the theoretical results, implemented the code, designed the experiments and significantly contributed to the writing.
- 2022 **Undergraduate Research** Worked on a research project about statistical inverse problems. I implemented Python versions of the proposed algorithms and proofread the paper manuscript. The final paper was published in [NIPS](#).

Teaching

- 2021 — Now **Teaching Assistant** Worked as teaching assistant for the following classes: Linear Algebra, Probability Theory, Real Analysis, Machine Learning (all undergraduate level) and Measure Theory (two times, graduate level). Duties generally included hosting tutoring sessions, as well as elaborating and grading homework assignments/tests.
- 2023 **FGV Ampla** Served as a tutor for FGV’s student-led college entrance exam preparation classes for low-income high school students.

Publications

- (Accepted at NeurIPS 2024) Y. Fonseca, **C. Peixoto**, Y. Saporito. Nonparametric instrumental variable regression through stochastic approximate gradients. arXiv:2402.05639 (2024).

Awards

- 2024 **CAPES & IMPA** (*Brazilian Federal Agency for Support and Evaluation of Graduate Education & National Institute for Pure and Applied Mathematics*) Awarded **one of 15** master’s level scholarships targeted towards Brazilian master’s students in Mathematics who had been medalists in Brazilian mathematical olympiads at some point in their studies.
- 2024 **SBMAC** Academic merit prize awarded by the *Brazilian Society for Applied and Computational Mathematics* (SBMAC) to “honor students which stand out through their academic performance in Applied and/or Computational Mathematics in STEM undergraduate courses”.

- 2023 **FGV EMAP** Academic merit prize (highest average grade of my cohort).
- 2017—19 **OBMEP** (*Brazilian Mathematical Olympiad for Public Schools*). Two Silver medals and one Bronze medal.

Event Participation

- 2024 **Organizer** 12th World Congress of the Bachelier Finance Society, FGV EMAP, Rio de Janeiro, Brazil. Member of the Local Committee.
- 2023 **Presenter** RiO — Research in Options, Rio de Janeiro, Brazil. Poster entitled “Nonparametric instrumental regression through stochastic approximate gradients”.
- 2023 **Attender** LACIAM — Latin American Congress on Industrial and Applied Mathematics.
- 2022 **Attender** CNMAC — National Congress of Applied and Computational Mathematics, Campinas, Brazil.

Skills

Programming Python (NumPy, Tensorflow, Jax, PyTorch), Julia, C++.

Software $\LaTeX$, Git, Linux, Vim.

Languages

Portuguese Native

English C1